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Quantum Error Correcting Subsystem Codes and Fault-Tolerant Quantum Computing

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Funded by:



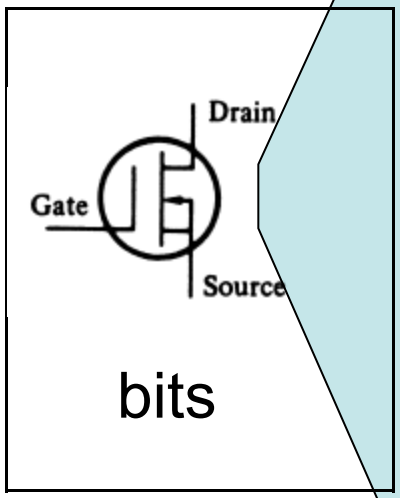
Talk Design Strategy

Ramping Up

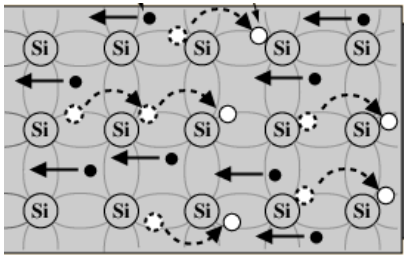
Bit, Pbit, and Qubit

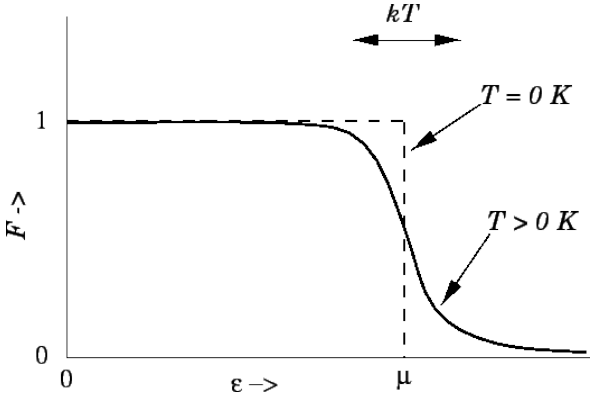
	bit	probabilistic bit	quantum bit
Configurations:	0 1	0 1	0 1
Description:	$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$	$\begin{bmatrix} p \\ 1-p \end{bmatrix}$ $p \in \mathbb{R}$	$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ $\alpha, \beta \in \mathbb{C}$
Observation:	0 certainty	0 *p* percent 1 *1-p* percent	0 $\lvert\alpha\rvert^2$ercent 1 $\lvert\beta\rvert^2$ercent
Evolution:	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	$\begin{bmatrix} 1-a & b \\ a & 1-b \end{bmatrix}$	$\begin{bmatrix} u & v \\ w & x \end{bmatrix}$
	deterministic	stochastic	unitary

There Are No Bits

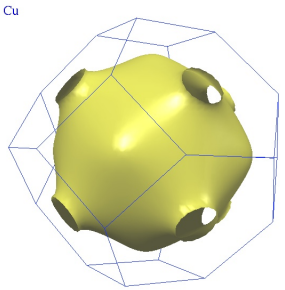


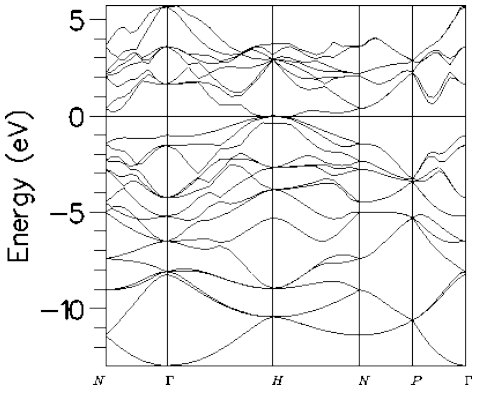
bits





pbits





qubits

Quantum Computers

188198812920607963838697239461650439807163563379417382700763356422988859715234665485319060606504743045317388011303396716199692321205734031879550656996221305168759307650257059

=

3980750864240649373971255005503864911990643623425267084063851895754638895726176858331747277214610743530253622307197304822463291469530209711645985217113052071125636359039752 7



Peter Shor

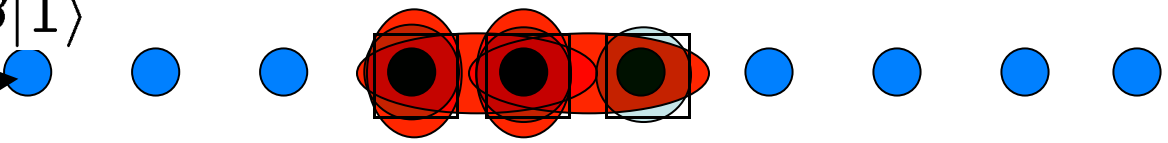
1994

- Quantum computers can efficiently factor whole numbers (and compute discrete log.)
- No known classical efficient algorithm.
- Breaks RSA, Diffie-Hellman cryptosystems.

Dreamboat Quantum Computer

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

qubits



$$\frac{1}{2}(|000\rangle + |011\rangle + |100\rangle - |111\rangle)$$

Quantum algorithm

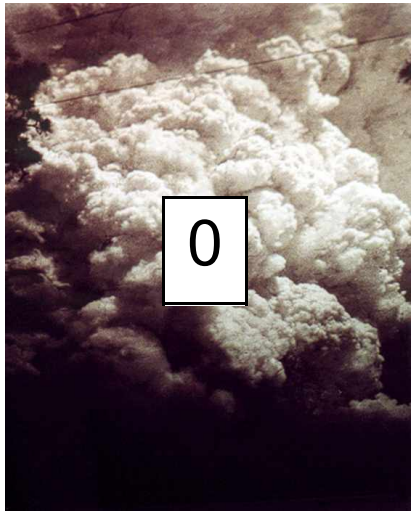
1. Initialize
2. Series of unitary gates
3. Measurement

0

1

1

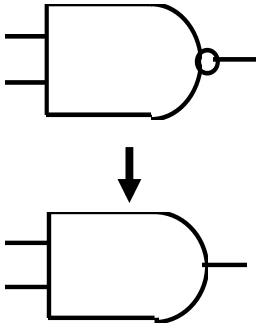
Fragile

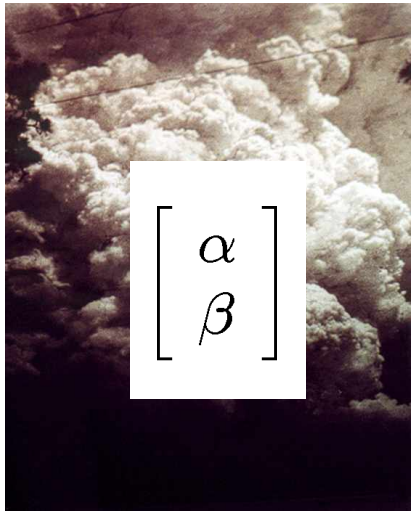


Classical Noise

$$\begin{bmatrix} 1-a & b \\ a & 1-b \end{bmatrix}$$

Classical Faults



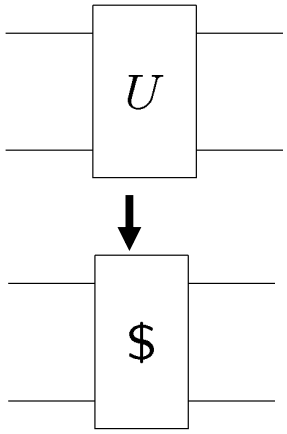


Quantum Noise

$$\begin{bmatrix} a_{00} & a_{01} \\ a_{10} & a_{11} \end{bmatrix}$$

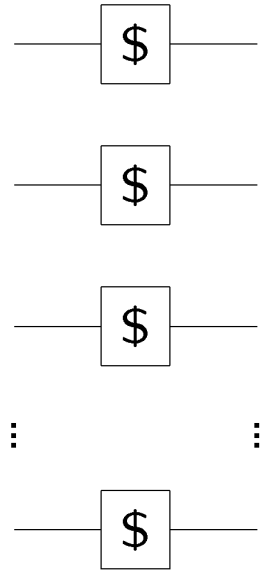
$$\begin{bmatrix} b_{00} & b_{01} \\ b_{10} & b_{11} \end{bmatrix}$$

Quantum Faults

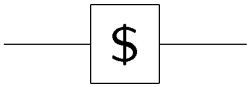


Reasonable Quantum Error Model

independent error model



basis for errors



$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$XZ = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

transition amplitude for each error

Reason's To Be Pessimistic, 1994

No cloning:



Quantum Cloning Machine

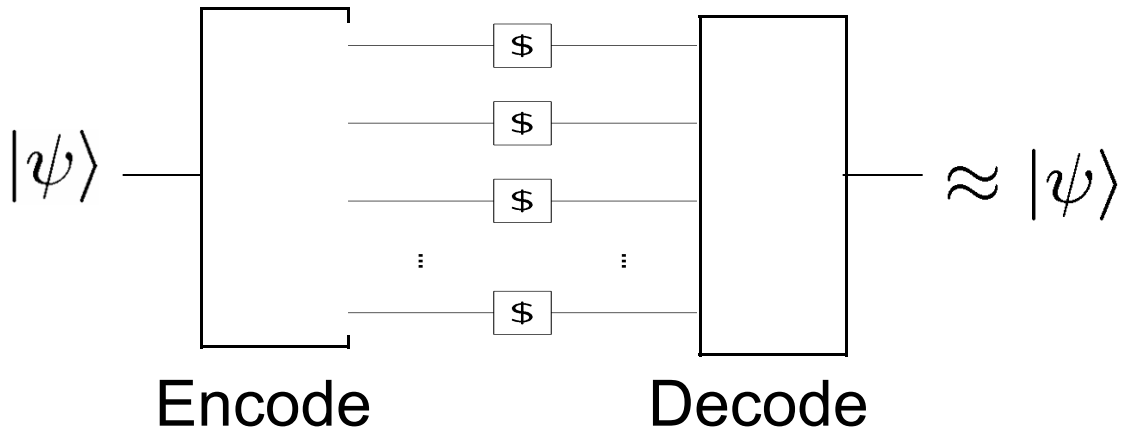
No quantum repetition code:

$$|\psi\rangle \nrightarrow |\psi\rangle \otimes |\psi\rangle \otimes |\psi\rangle$$

Measurement destroys coherence:

How can one decode without destroying the information?

Reason's To Be Optimistic, 1994



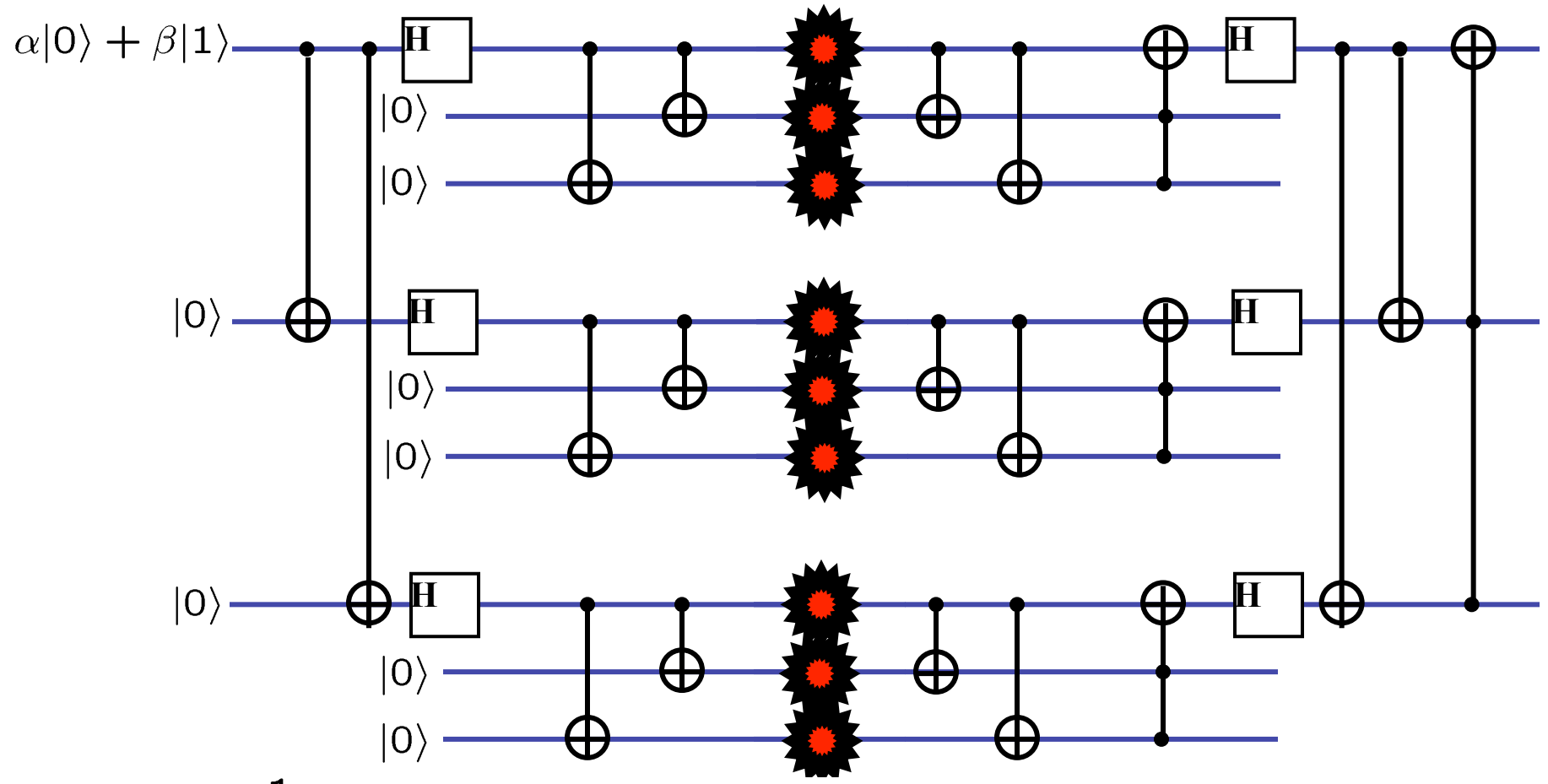
$$\begin{bmatrix} \alpha \\ \beta \end{bmatrix} \qquad \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

$|\psi\rangle \in \mathbb{C}^2$ $\qquad$ $|\psi\rangle \in \mathcal{H} \subset \mathbb{C}^{2n}$ $\qquad$ $|\psi\rangle \in \mathbb{C}^2$

Encode into a *subspace* of n qubits

Possible to design encoding/decoding such that any single qubit error can be corrected: *quantum error correcting code*

Shor's Code



$$|0\rangle \rightarrow \frac{1}{2\sqrt{2}}(|000\rangle+|111\rangle)\otimes(|000\rangle+|111\rangle)\otimes(|000\rangle+|111\rangle)$$

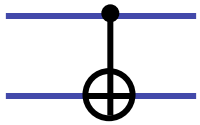
$$|1\rangle \rightarrow \frac{1}{2\sqrt{2}}(|000\rangle-|111\rangle)\otimes(|000\rangle-|111\rangle)\otimes(|000\rangle-|111\rangle)$$

A Start

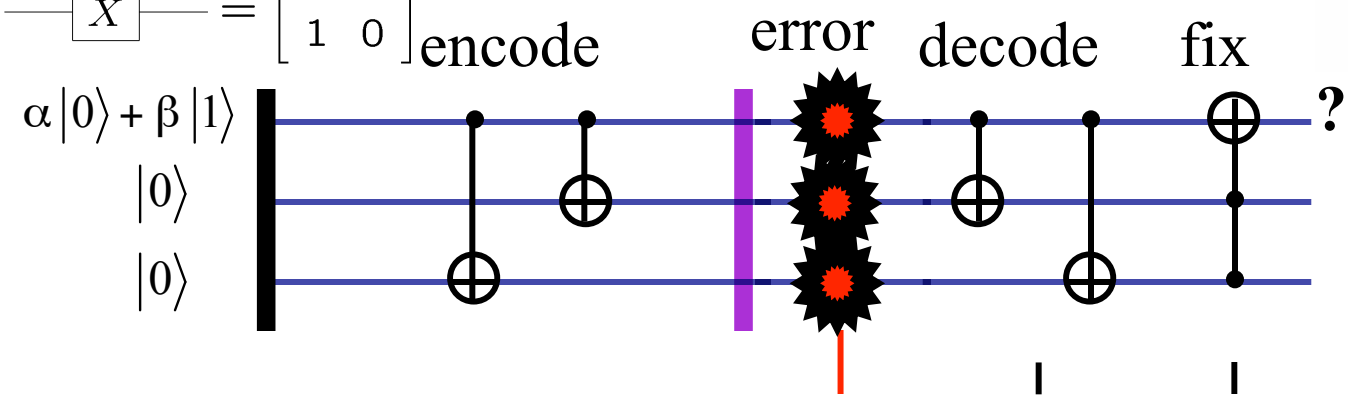


$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$



$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$



$$(\alpha|0\rangle + \beta|1\rangle) \otimes |0\rangle \otimes |0\rangle \xrightarrow{U} \alpha|0\rangle \otimes |0\rangle \otimes |0\rangle + \beta|1\rangle \otimes |1\rangle \otimes |1\rangle$$

$$\alpha|000\rangle + \beta|111\rangle \overset{XII}{\rightarrow} \alpha|100\rangle + \beta|011\rangle \rightarrow (\alpha|1\rangle + \beta|0\rangle)|11\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)|11\rangle$$

$$\alpha|000\rangle + \beta|111\rangle \overset{IXI}{\rightarrow} \alpha|010\rangle + \beta|101\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)|10\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)|10\rangle$$

$$\alpha|000\rangle + \beta|111\rangle \overset{IIX}{\rightarrow} \alpha|001\rangle + \beta|110\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)|01\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)|01\rangle$$

error decode fix

encoded into subspace:

$|0\rangle \rightarrow |000\rangle, |1\rangle \rightarrow |111\rangle$

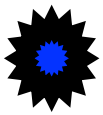
(no-cloning evaded!)

2. errors take to orthogonal subspaces + maintain orthogonality

$\mathbf{XII}|000\rangle = |100\rangle, \mathbf{XII}|111\rangle = |011\rangle$

3. syndrome

OK Wise Guy



$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$Z|0\rangle = |0\rangle$$
$$Z|1\rangle = -|1\rangle$$

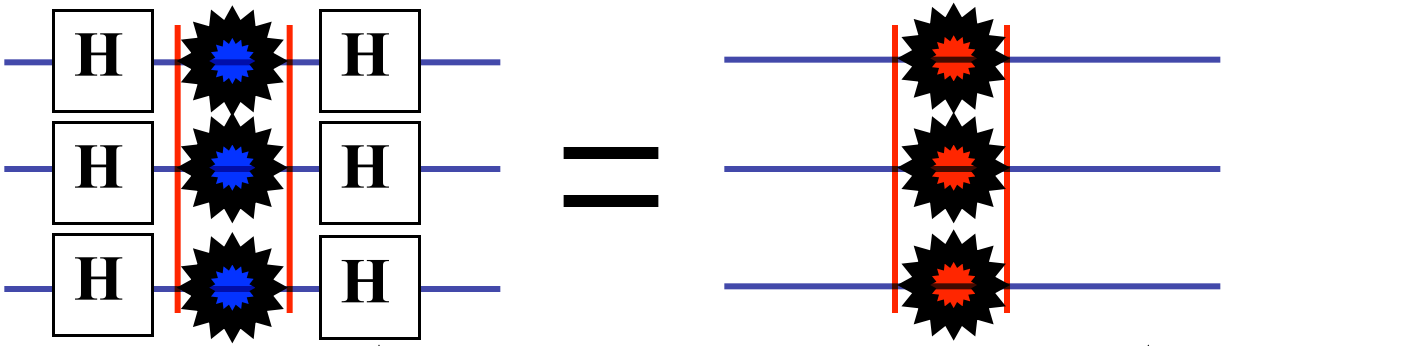
Wise guy says "basis change please":

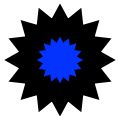
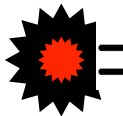
$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$
$$|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$$Z|+\rangle = |-\rangle$$
$$Z|-\rangle = |+\rangle$$

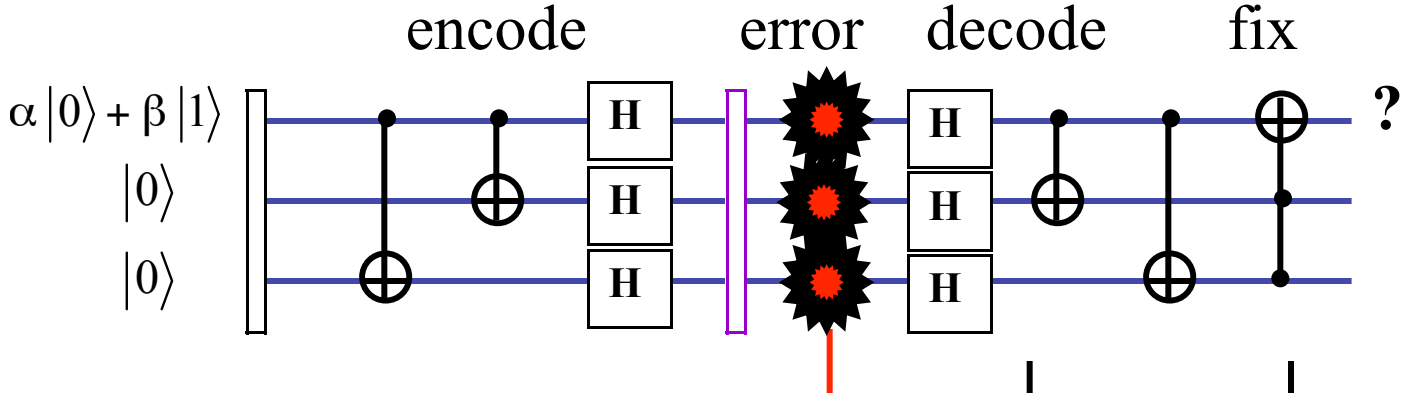
looks like bit flit error in this new basis!

$$HZH = X, \quad H = |+\rangle\langle 0| + |-\rangle\langle 1|$$



phase errors  $= Z$ bit flip errors  $= X$

Molly: “I love you, I really love you”
Sam: “Ditto.”



$$(\alpha|0\rangle + \beta|1\rangle) \otimes |0\rangle \otimes |0\rangle \xrightarrow{U} \alpha|+\rangle \otimes |+\rangle \otimes |+\rangle + \beta|-\rangle \otimes |-\rangle \otimes |-\rangle$$

$$\alpha|+++\rangle + \beta|---\rangle \xrightarrow{ZII} \alpha|-++\rangle + \beta|+--\rangle \rightarrow (\alpha|1\rangle + \beta|0\rangle)\,|11\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)\,|11\rangle$$

$$\alpha|+++\rangle + \beta|---\rangle \xrightarrow{IZI} \alpha|+-+\rangle + \beta|-+-\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)\,|10\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)\,|10\rangle$$

$$\alpha|+++\rangle + \beta|---\rangle \xrightarrow{IIZ} \alpha|++-\rangle + \beta|--+\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)\,|01\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)\,|01\rangle$$

error decode fix

encoded into subspace:

$|0\rangle \rightarrow |+++\rangle, |1\rangle \rightarrow |---\rangle$

(no-cloning evaded!)

2. errors take to orthogonal subspaces + maintain orthogonality

$\mathbf{ZII}|+++\rangle = |-++\rangle, \mathbf{ZII}|---\rangle = |+--\rangle$

3. syndrome

Encoding Away Your Ills

3 qubit bit flip code	3 qubit phase flip code
$\|0\rangle \rightarrow \|000\rangle$	$\|0\rangle \rightarrow \|+++\rangle$
$\|1\rangle \rightarrow \|111\rangle$	$\|1\rangle \rightarrow \|---\rangle$

phase errors **ZII, IZI, IIZ** act as ■on bit flip code qubits:

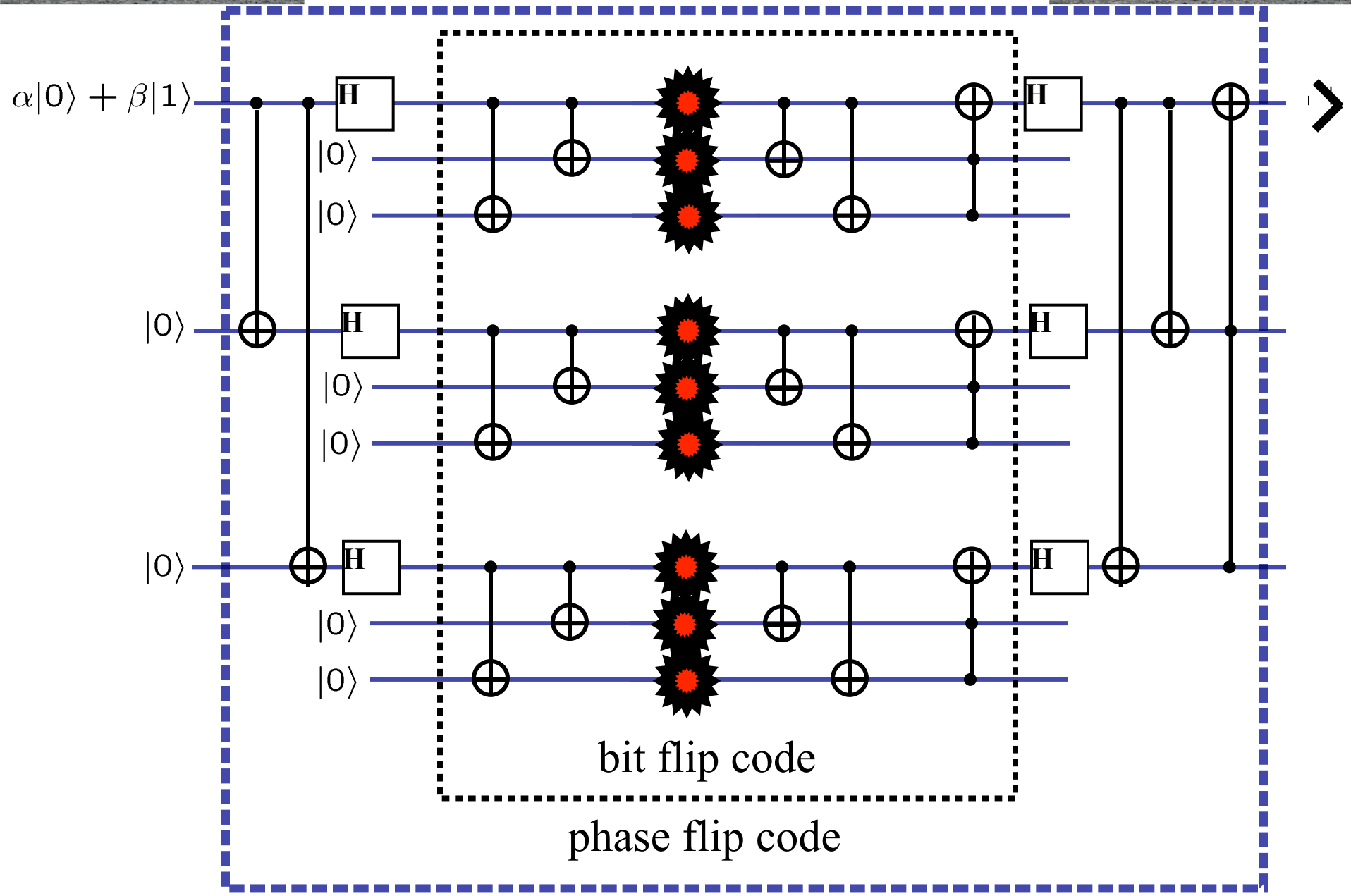
$$|0\rangle_B = |000\rangle \qquad \mathbf{ZII}|0\rangle_B = |0\rangle_B$$
$$|1\rangle_B = |111\rangle \qquad \mathbf{ZII}|1\rangle_B = -|1\rangle_B$$

define:
$$|p\rangle = \frac{1}{\sqrt{2}}(|0\rangle_B + |1\rangle_B) = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$$
$$|m\rangle = \frac{1}{\sqrt{2}}(|0\rangle_B - |1\rangle_B) = \frac{1}{\sqrt{2}}(|000\rangle - |111\rangle)$$

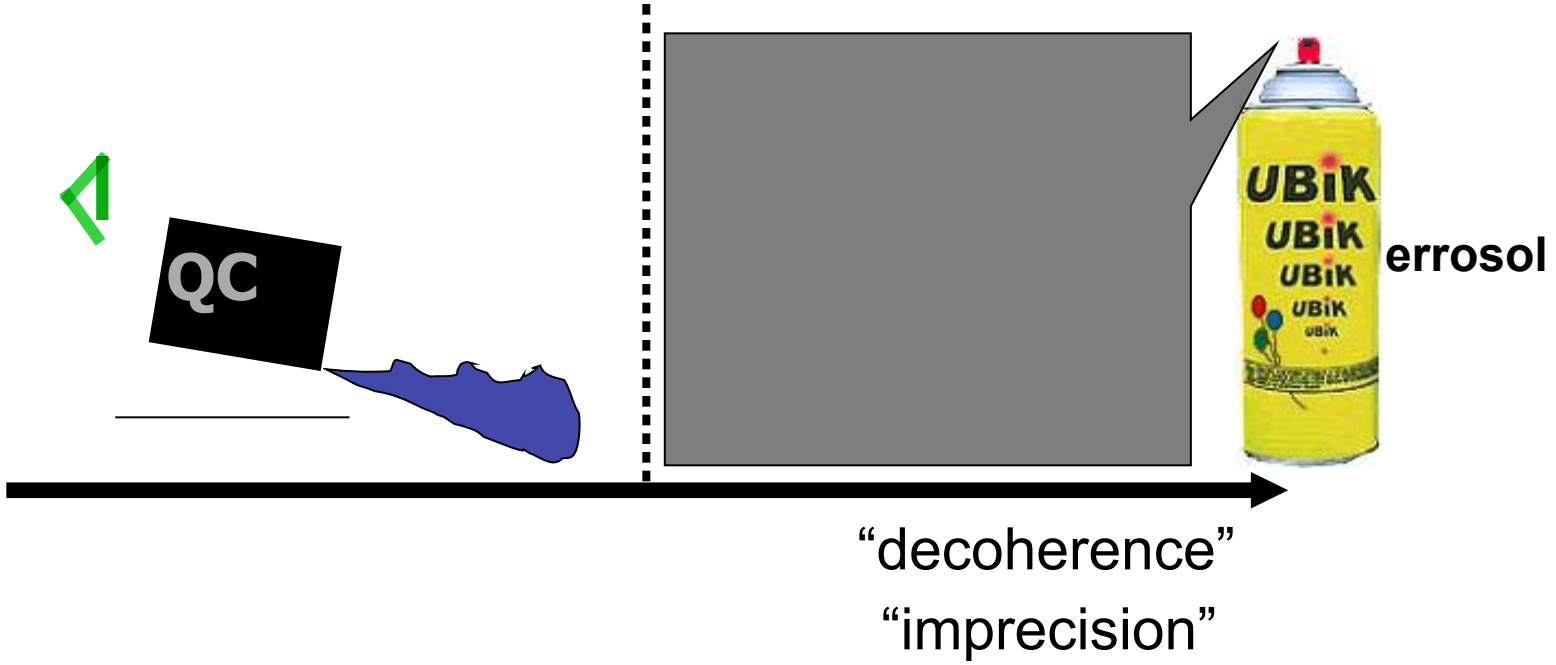
Shor Code: (Peter Shor, 1995)

$$|0\rangle \rightarrow |ppp\rangle = \frac{1}{2\sqrt{2}}(|000\rangle + |111\rangle)(|000\rangle + |111\rangle)(|000\rangle + |111\rangle)$$
$$|1\rangle \rightarrow |mmm\rangle = \frac{1}{2\sqrt{2}}(|000\rangle - |111\rangle)(|000\rangle - |111\rangle)(|000\rangle - |111\rangle)$$

Shor's Code


$\alpha|0\rangle + \beta|1\rangle$
$|0\rangle$
$|0\rangle$
H
H
H
H
H
H
bit flip code
phase flip code

Seminal Result



Threshold Theorem for Fault-Tolerant Quantum Computation

[a partial list: Aharonov, Ben-Or, Kitaev, Knill, LaFlamme, Zurek, Shor, DiVincenzo, Gottesman, Preskill]

If decoherence (quantum noise) is weak enough and control is precise enough,then a quantum computation to a desired accuracy can be carried out with an overhead polynomial in the log of the accuracy.

Threshold Theorems

TWO TYPES

provable and heuristic

ASSUMPTIONS

provably essential:

1. parallel operations
2. cold ancillas
3. error rates below 50% gate speed
4. ability to not leak or fix leaks

error model:

correlations (spatial and temporal)

quantum noise (single qubit, leakage, Markovian, etc.)

classical computation:

free fast robust classical computation?

architecture:

non-local gates, local gates

dimension of architecture

explicit architecture (scheduling, routing, etc.)

→Thresholds from 10-1 to 10-8

Codes Better For Fault-Tolerance?

Subspaces Versus Subsystems

The most general way to encode quantum information is into subsystem (not subspaces)



two qubits

$$\mathcal{H}_2 \otimes \mathcal{H}_2$$

<u>encode a qubit in a subspace</u>

$$|0_L\rangle = |1\rangle \otimes |1\rangle$$

$$|1_L\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |1\rangle - |1\rangle \otimes |0\rangle)$$

$$\alpha|0_L\rangle + \beta|1_L\rangle$$

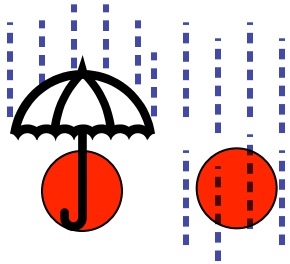
<u>encode a qubit in a subsystem</u>

$$(\alpha|0_L\rangle + \beta|1_L\rangle) \otimes |\psi\rangle$$

generic subsystem:

$$\mathcal{H}_{sys} = (\mathcal{C} \otimes \mathcal{D}) \oplus \mathcal{P}$$

Subsystem Error Correction



Only need to protect subsystem, not full subspaces.

$$\mathcal{H}_{sys} = (\mathcal{C} \otimes \mathcal{D}) \oplus \mathcal{P}$$

encoding subsystem

$(|\phi\rangle\langle\phi| \otimes \rho_D) \oplus 0$ — $\mathcal{E}$ — $\mathcal{R}$ — $(|\phi\rangle\langle\phi| \otimes \tilde{\rho}_D) \oplus 0$

error recovery

encoded quantum information

recovery need not perfectly restore system, but only preserve information encoded into subsystem.

Shor Stabilizer

Quantum version of parity check matrix

$$|0_L\rangle = \frac{1}{2\sqrt{2}}(|000\rangle+|111\rangle)\otimes(|000\rangle+|111\rangle)\otimes(|000\rangle+|111\rangle)$$

$$|1_L\rangle = \frac{1}{2\sqrt{2}}(|000\rangle-|111\rangle)\otimes(|000\rangle-|111\rangle)\otimes(|000\rangle-|111\rangle)$$

Stabilizer generators:

$$ZZIIIIIII \quad IIIZZIIII \quad IIIIIIZZI \quad XXXXXXIII$$
$$IZZIIIIII \quad IIIIZZIII \quad IIIIIIIZZ \quad IIIXXXXXX$$

Define subspace:

$$S|\psi\rangle = |\psi\rangle$$

Logical operators:

$$\bar{Z} = ZZZZZZZZZ \qquad \bar{Z}|0_L\rangle = |0\rangle$$
$$\bar{X} = XXXXXXXXX \qquad \bar{Z}|1_L\rangle = -|1\rangle$$

Shor Redux

Stabilizer generators:

$$\begin{matrix} X & X & X \\ X & X & X \\ I & I & I \end{matrix} \qquad \begin{matrix} I & I & I \\ X & X & X \\ X & X & X \end{matrix} \qquad \begin{matrix} Z & Z & I \\ Z & Z & I \\ Z & Z & I \end{matrix} \qquad \begin{matrix} I & Z & Z \\ I & Z & Z \\ I & Z & Z \end{matrix}$$

Encodes 9-4=5 qubits of information

Logical operators:

$$\bar{X}_1 = \begin{matrix} X & I & I \\ X & I & I \\ I & I & I \end{matrix} \quad \bar{X}_2 = \begin{matrix} I & I & X \\ I & I & X \\ I & I & I \end{matrix} \quad \bar{X}_3 = \begin{matrix} I & I & I \\ X & I & I \\ X & I & I \end{matrix} \quad \bar{X}_4 = \begin{matrix} I & I & I \\ I & I & X \\ I & I & X \end{matrix}$$

$$\bar{Z}_1 = \begin{matrix} Z & Z & I \\ I & I & I \\ I & I & I \end{matrix} \quad \bar{Z}_2 = \begin{matrix} I & Z & Z \\ I & I & I \\ I & I & I \end{matrix} \quad \bar{Z}_3 = \begin{matrix} I & I & I \\ I & I & I \\ Z & Z & I \end{matrix} \quad \bar{Z}_4 = \begin{matrix} I & I & I \\ I & I & I \\ I & Z & Z \end{matrix}$$

$$\bar{X}_5 = \begin{matrix} X & X & X \\ X & X & X \\ X & X & X \end{matrix} \quad \bar{Z}_5 = \begin{matrix} Z & Z & Z \\ Z & Z & Z \\ Z & Z & Z \end{matrix}$$

[Bacon, Phys. Rev A, 73, 012340]

Seen This Before

Measuring +1,-1 eigenvalue of

$$\begin{matrix} X & X & X \\ I & I & I \\ I & I & I \end{matrix} \qquad \begin{matrix} I & I & I \\ X & X & X \\ I & I & I \end{matrix} \qquad \begin{matrix} I & I & I \\ I & I & I \\ X & X & X \end{matrix}$$

determines whether the number of phase flips in each row is even or odd

Measuring +1,-1 eigenvalue of

$$\begin{matrix} X & X & X \\ X & X & X \\ I & I & I \end{matrix} \qquad \begin{matrix} I & I & I \\ X & X & X \\ X & X & X \end{matrix}$$

is just like a simple redundancy code for this even/odd info

[Bacon, Phys. Rev A, 73, 012340]

Subsystem Code

Stabilizer generators:

$$\begin{matrix} Z & Z & I & I & Z & Z \\ Z & Z & I & I & Z & Z \\ Z & Z & I & I & Z & Z \end{matrix}$$

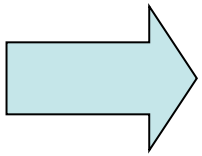
plus encoded Z operators

$$\bar{Z}_1 = \begin{matrix} Z & Z & I \\ I & I & I \\ I & I & I \end{matrix} \quad \bar{Z}_2 = \begin{matrix} I & Z & Z \\ I & I & I \\ I & I & I \end{matrix} \quad \bar{Z}_3 = \begin{matrix} I & I & I \\ I & I & I \\ Z & Z & I \end{matrix} \quad \bar{Z}_4 = \begin{matrix} I & I & I \\ I & I & I \\ I & Z & Z \end{matrix}$$

generate all even number of Z's per row

Measuring +1,-1 eigenvalue of

$$\begin{matrix} X & X & X & I & I & I \\ X & X & X & X & X & X \\ I & I & I & X & X & X \end{matrix}$$

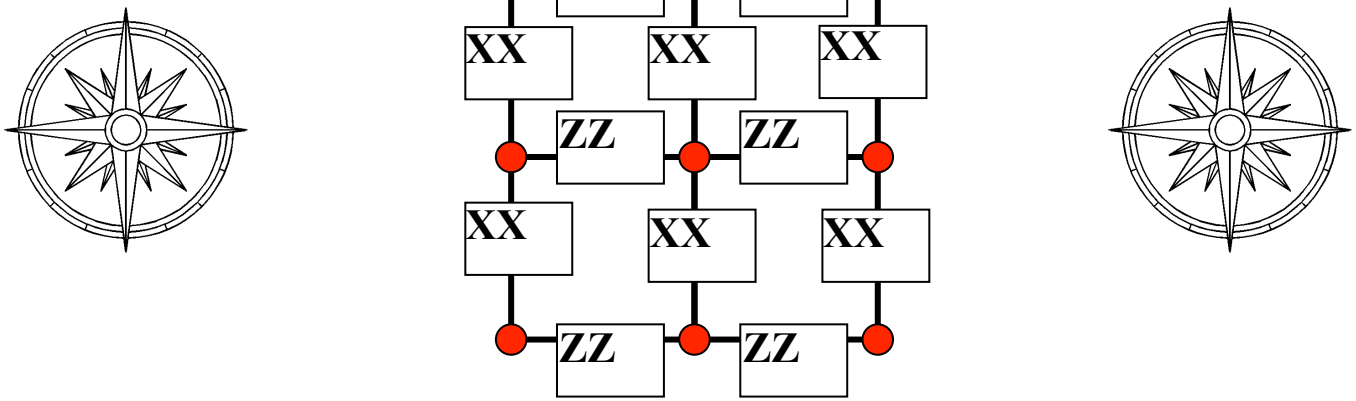


Fix modulo this group
i.e. do nothing to 5th encoded qubit!

[Bacon, Phys. Rev A, 73, 012340]

Quantum Compass Model

$$H = -J_X \sum_{ij} X_{i,j} X_{i+1,j} - J_Z \sum_{ij} Z_{i,j} Z_{i,j+1}$$



Was studying because I wanted to build a naturally fault-tolerant quantum system (physics enforces robustness.)

But had implications for fault-tolerance!

Aliferis and Cross, in prep (2006, soon!), have shown that these codes can move threshold a factor of 10 better.

Generalized Shor Subsystem Codes

Two classical codes: C1=[n1,k1,d1] and C2=[n2,k2,d2]

Quantum Shor subspace code [[n1n2,k1k2,min(d1,d2)]]
Number of stabilizers = (n1-k1)n2+(n2-k2)

Quantum Shor subsystem code [[n1n2,k1k2,min(d1,d2)]]
Number of stabilizers= (n1-k1)k2+(n2-k2)k1

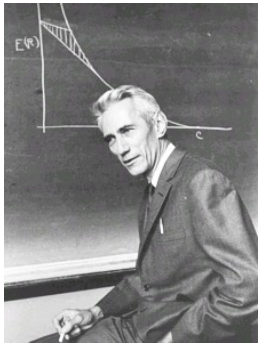
Same exciting ancilla savings as
Aliferis and Cross, in prep (2006)

Shor Subsystem Codes

Simpler error recovery routines

Better fault-tolerant properties

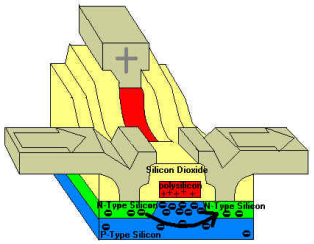
Useful for physicists in constructing physically robust qubits





theory





practice

Practical devices of information processing enact the ideas of fault-tolerant classical computation in the *physics* of the devices.

Quantum Error Correcting Subsystem Codes and Fault-Tolerant Quantum Computing

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